

AIDOS LAB
AI FOR DATA-ORIENTED SCIENCE

UNI FR

Lecture 8: (More) persistent homology

Geometry and Topology in Machine Learning

Inés García Redondo — November 3, 2026

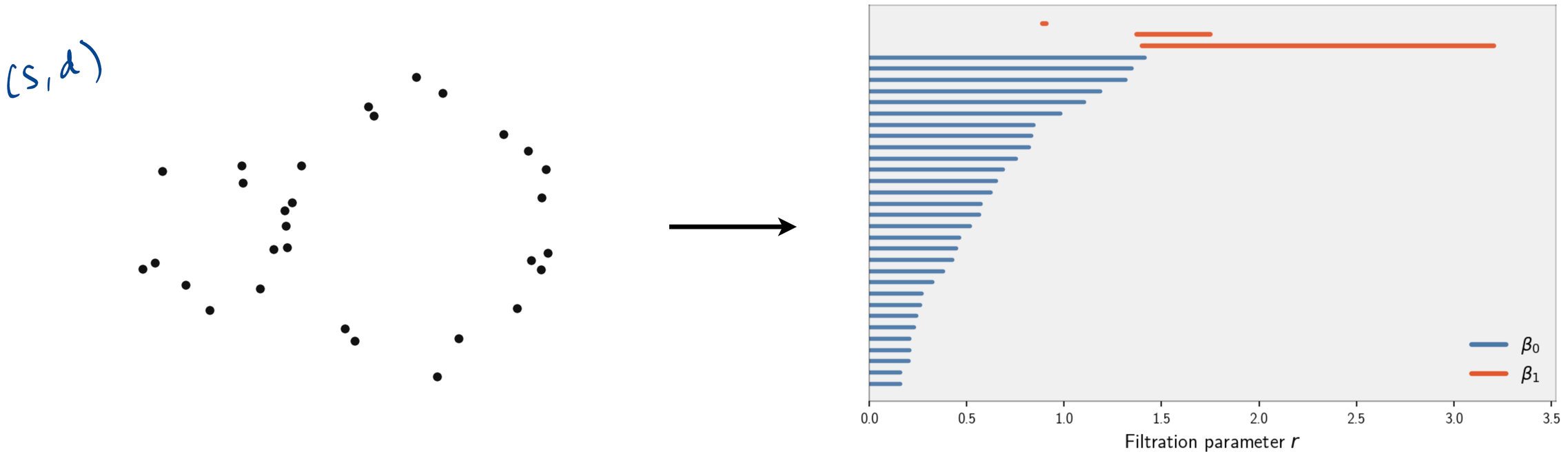
Recap

Lecture 8 — (More) persistent homology

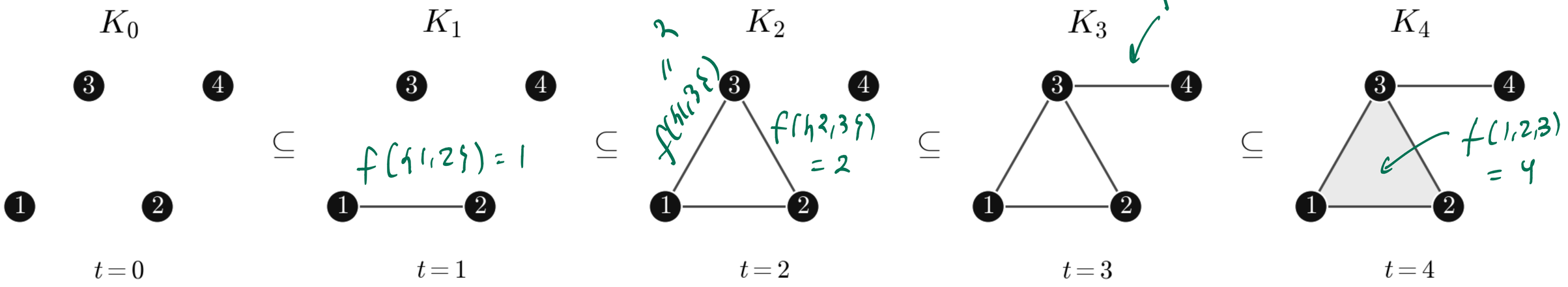
Persistence barcode

Last week we saw that for every input data set coming as a finite metric space (think of a point cloud) we could compute a persistence barcode

This persistence barcode is obtained choosing an appropriate basis for the homology groups at each step in the filtration and tracking where the maps induced by the inclusions of the filtration take the basis elements



Filtrations



Filtration: $\{K_t : 0 \le t \le N\}$, s.t. for $s \le t$, $K_s \subseteq K_t$

We can think of ir as being obtained from a *monotonic function* on the last simplicial complex $K = K_N$

Def. Monotonic simplicial function

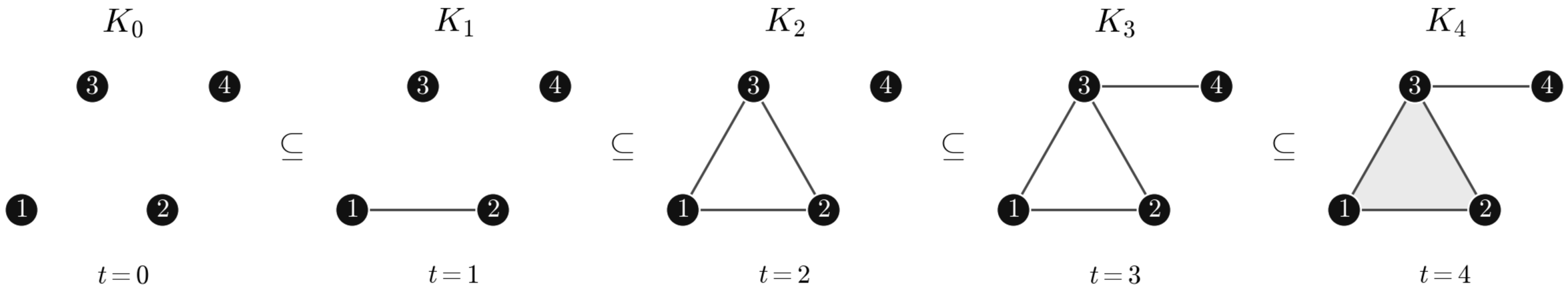
Let K be a simplicial complex. A function $f : K \to \mathbb{R}$ is monotonic iff for all $\sigma \subseteq \tau \in K$, $f(\sigma) \le f(\tau)$

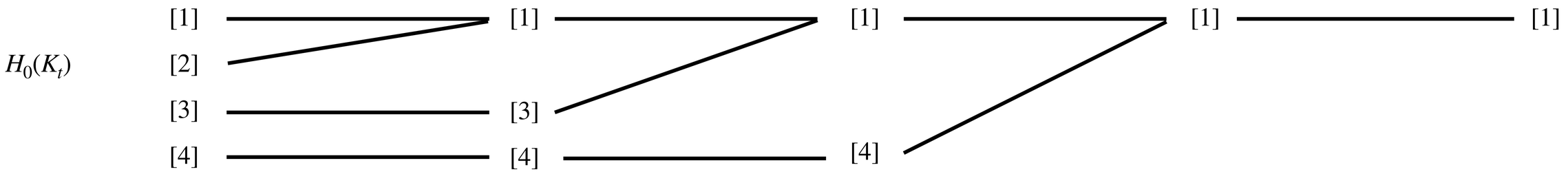
With such a function we can get a filtration by setting:

$$K_t = f^{-1}(-\infty, t] = \{\sigma \in K : f(\sigma) \le t\}$$

Sublevel set filtration

Getting the barcode



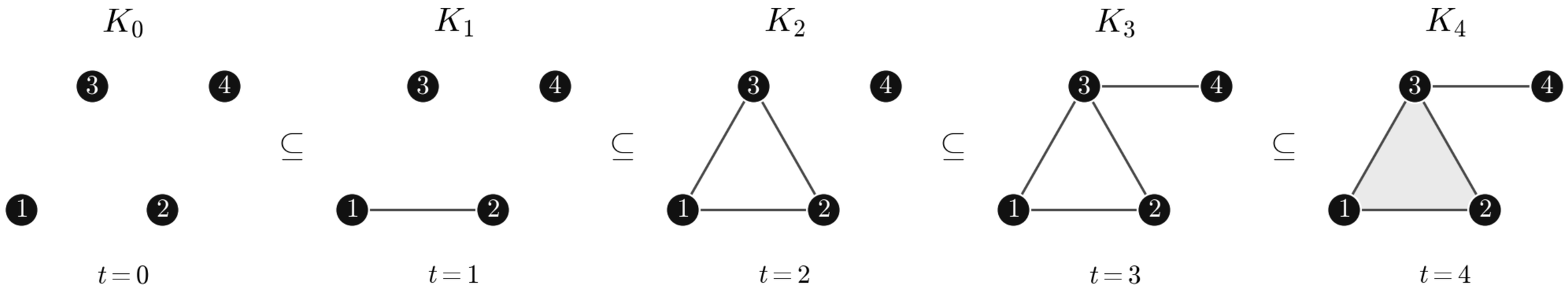


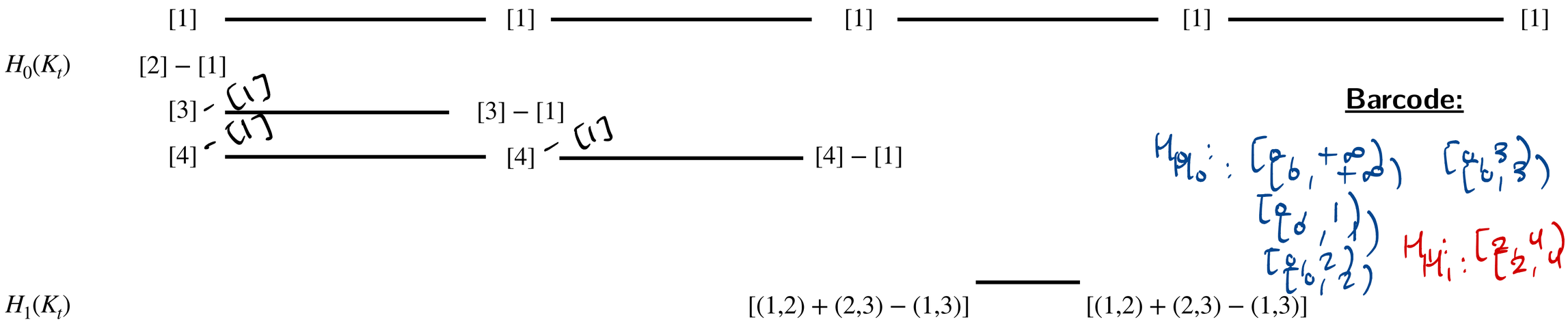
$H_1(K_t)$

[(1,2) + (2,3) − (1,3)] —— [(1,2) + (2,3) − (1,3)]

Getting the barcode

If I choose my generators more wisely...





Barcode:

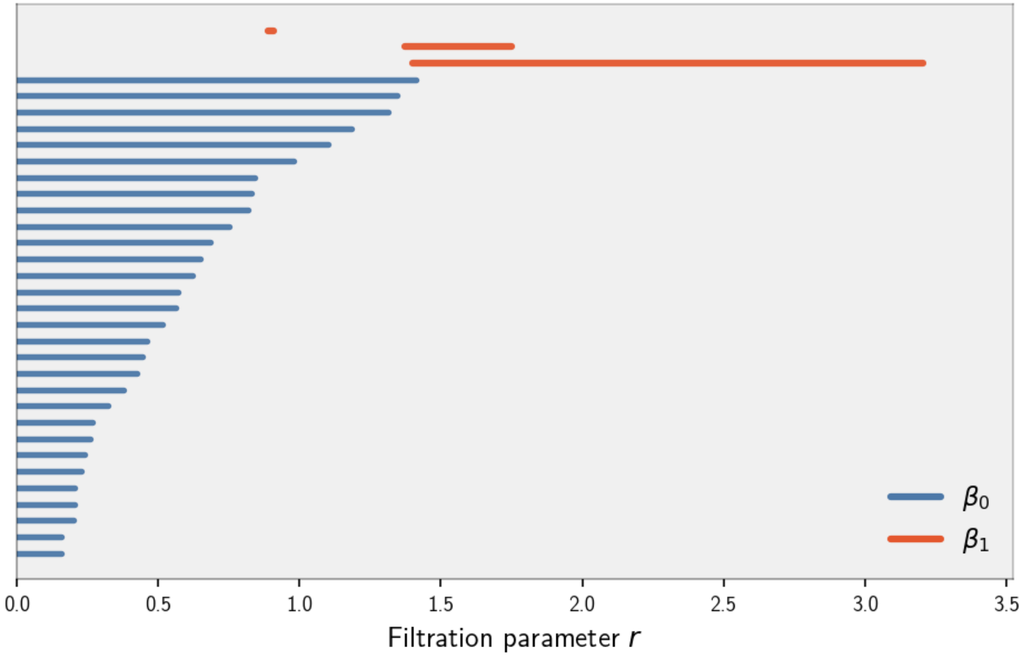
H_0: $[0, +\infty)$ $[0, 3)$ $[0, 1)$ $[0, 2)$

H_1: $[2, 4)$

Persistence barcode



As many bars as points, bars *die* as points get connected

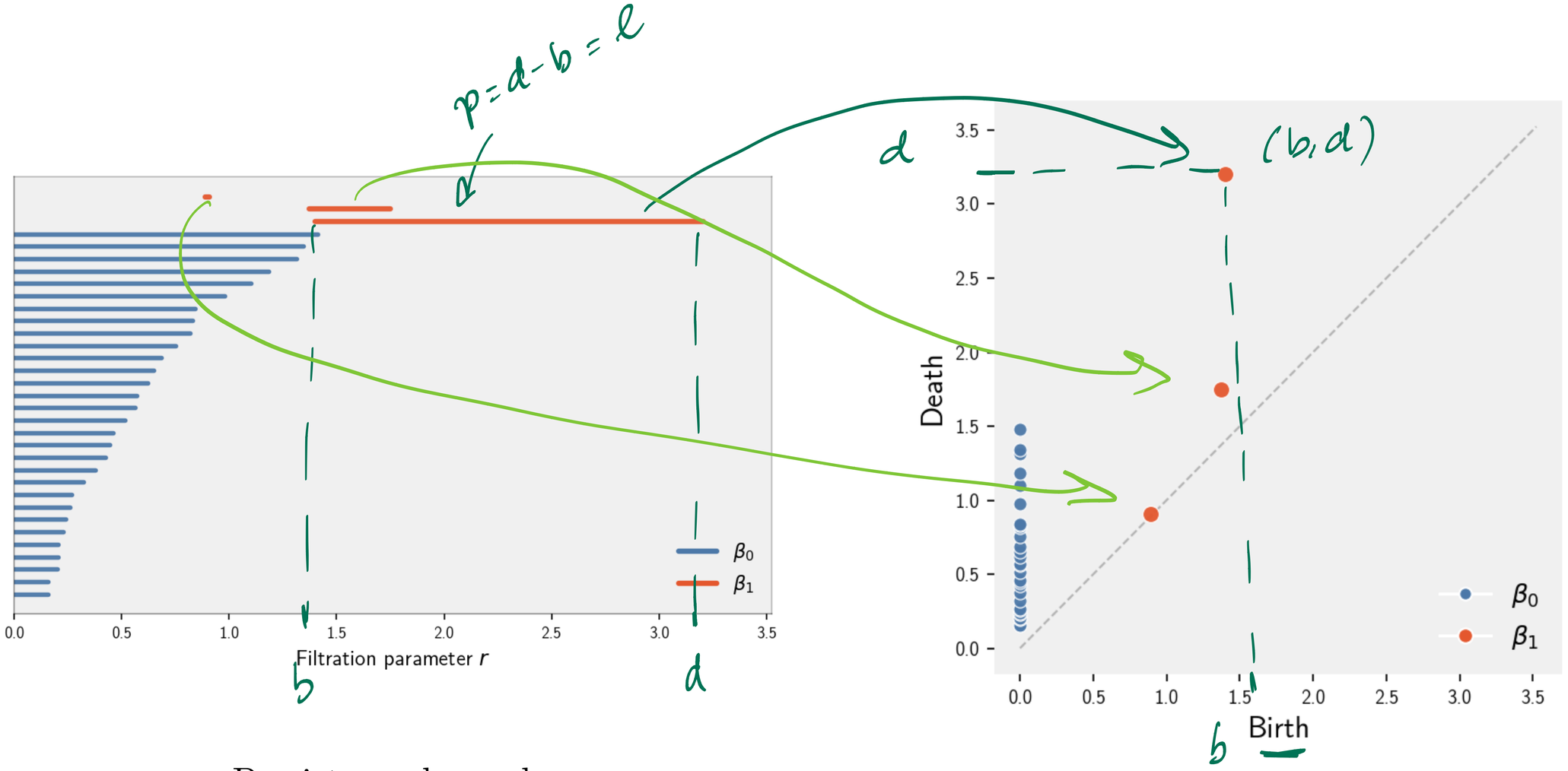


Two long bars: we have two *persistent* loops in the point cloud

The point cloud had the shape of two circles!

Persistence barcode

Persistence diagram



Persistence barcode

Persistence diagram

Computing barcodes

Lecture 8 — (More) persistent homology

Filtration boundary matrix

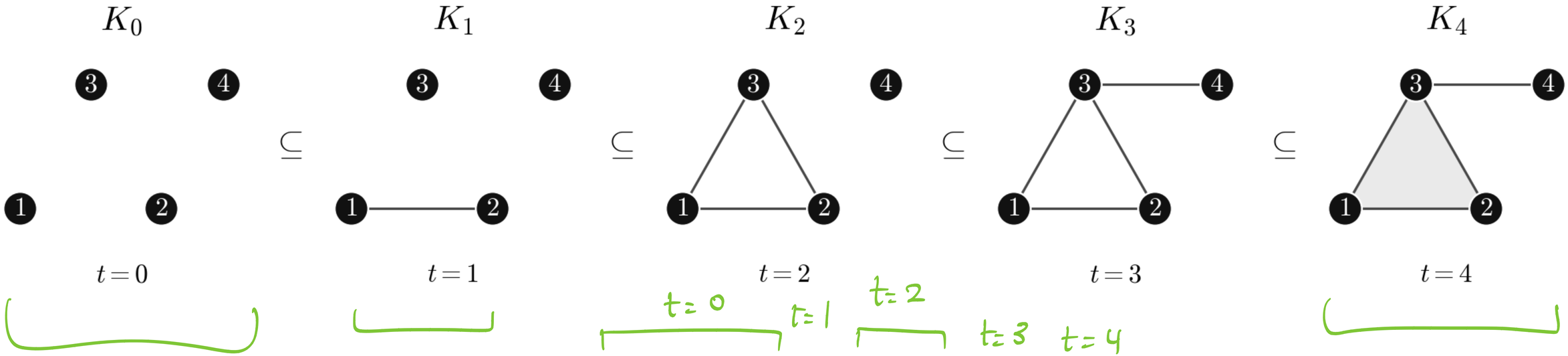
How to define it?

- Order all simplices in the simplicial complex at the last step of the filtration $K = K_N$ by increasing filtration value
- When several simplices enter the filtration at the same time, order them by dimension
- Break ties arbitrarily when two simplices of the same dimension enter the filtration at the same time
- Compute the boundary matrices D_k representing the boundary maps $\partial_k : C_k(K) \to C_{k-1}(K)$ of the simplicial complex at the last step of the filtration, where all simplices have been added
- Write the full boundary matrix as the block matrix given by all these matrices

$$D = \bigoplus_{k=0}^{\dim K} D_k$$

- D_{ij} = coefficient of σ_i in $\partial\sigma_j$ with indices ranging all the simplices in K

Reducing the boundary matrix



$t=0$ $t=1$ $t=2$ $t=3$ $t=4$

$$D = \left(\begin{array}{c|cccc|ccccc} & 1 & 2 & 3 & 4 & 12 & 13 & 23 & 34 & 123 \\ \hline 1 & 0 & 0 & 0 & 0 & -1 & -1 & 0 & 0 & 0 \\ 2 & 0 & 0 & 0 & 0 & 1 & 0 & -1 & 0 & 0 \\ 3 & 0 & 0 & 0 & 0 & 0 & 1 & 1 & -1 & 0 \\ 4 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ \hline 12 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \\ 13 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -1 \\ 23 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \\ 34 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ \hline 123 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{array}\right)$$

$\partial_2(1,2,3) =$

$(2,3) - (1,3) + (1,2)$

Reducing the boundary matrix

Reduction algorithm

Input: boundary matrix of a filtration D

Steps:

$R \leftarrow D$

$n \leftarrow$ # columns of R

For $1 \leq j \leq n$:

  If $R_j \neq 0$:

    pivot$(R_j) \leftarrow \max\{i : R_{ij} \neq 0\}$

    For $1 \leq k < j$:

      If pivot$(R_k) =$ pivot(R_j):

        $R_j \leftarrow R_j \pm R_k$ so that pivot(R_j) increases

Output: reduced matrix R

Reducing the boundary matrix

Reduction algorithm

$D =$

	1	2	3	4	12	13	23	34	123
1	0	0	0	0	−1	−1	0	0	0
2	0	0	0	0	1	0	−1	0	0
3	0	0	0	0	0	1	1	−1	0
4	0	0	0	0	0	0	0	1	0
12	0	0	0	0	0	0	0	0	1
13	0	0	0	0	0	0	0	0	−1
23	0	0	0	0	0	0	0	0	1
34	0	0	0	0	0	0	0	0	0
123	0	0	0	0	0	0	0	0	0

$R_{23} - R_{13}$

$\sim$

23
1
−1
0
0
0
0
0
0
0

$R_{23} + R_{12}$

$\sim$

23
0
0
0
0
0
0
0
0
0

$R = D$ with $R_{23} = 0$

Reading off the barcode

- If pivot$(R_j) = i$: σ_i creates a homology class and σ_j kills it. We have to add a *finite bar*

$$[f(\sigma_i), f(\sigma_j))$$

  where $f(\sigma_i)$ = filtration value at which σ_i enters the filtration, in dimension $\dim(\sigma_i)$

- If $R_j = 0$, that means that this column represents a simplex that creates a cycle. There are two options for that bar:
  - There exists a later column R_k, $j < k$ such that pivot$(R_k) = j$. Then the cycle created by σ_j is killed by σ_k and we have to add

$$[f(\sigma_j), f(\sigma_k))$$

    in dimension $\dim(\sigma_j)$

  - Otherwise, the cycle created by σ_j never gets killed, so we get an *essential bar*

$$[f(\sigma_j), +\infty)$$

    in dimension $\dim(\sigma_j)$

Positive and negative simplices

A bar $[f(\sigma_i), f(\sigma_j))$ obtained as explained in the previous slide corresponds to a topological feature that was created when σ_i was added and destroyed when σ_j was added

σ_i is called a **positive simplex**

σ_j is called a **negative simplex**

Bar		**Simplicial pairing**
$[f(\sigma_i), f(\sigma_j))$	$\longleftrightarrow$	$[\sigma_i, \sigma_j]$

Reading off the barcode

Example Find the barcode and the pairs of positive/negative simplices that generate each bar

$R =$

	1	2	3	4	12	13	23	34	123
1	0	0	0	0	−1	−1	0	0	0
2	0	0	0	0	1	0	0	0	0
3	0	0	0	0	0	1	0	−1	0
4	0	0	0	0	0	0	0	1	0
12	0	0	0	0	0	0	0	0	1
13	0	0	0	0	0	0	0	0	−1
23	0	0	0	0	0	0	0	0	1
34	0	0	0	0	0	0	0	0	0
123	0	0	0	0	0	0	0	0	0

$t=0$ $t=1$ $t=2$ $t=3$ $t=4$

H_0 : $[0, +\infty)$ $[0,3)$

$[0,1)$

$[0,2)$

$H_1 = [2, 4)$

H_0 : $[0, +\infty)$ $[0,3)$

$[0,1)$

$[0,2)$ H_1: $[2,4)$

The idea behind the algorithm

Notice that we are doing exactly the same thing as we did before:

- Starting from the basis of simplices in each dimension (basis of the chain space)
- We take the matrix representation of the boundary matrix in that basis
- We add of subtract simplices that have already appeared in the filtration to reduce the boundary, that is, *we change the element of the basis over which we are computing the boundary for the matrix representation*, adding the vectors of the basis that we already had
- We do this until we obtain a *boundary that is linearly independent* with respect to the other boundaries already present, or until we arrive to the zero column (that is, a combination of simplices which forms a *cycle*)
- This process gives us a basis from which we can read the barcode, just as before!

Complexity of the algorithm

- If the complex has n simplices, the matrix D has size $n \times n$
- When we are reducing a column, it can happen that rows that were 0 in the column turn to 1 in an operation
- In the words case, we need to perform n operations to cancel a columns $\rightsquigarrow$ $O(n)$
- Adding to columns with n entries adds a cost of $O(n)$ $\rightsquigarrow$ $O(n^2)$
- We need to repeat this for every column $\rightsquigarrow O(n^3)$

- There has been a lot of working in making this algorithm more efficient in terms of *storage* (avoiding to store the boundary matrix which becomes huge)
  - Use for example the sparsity of the boundary matrix (a d-simplex has $d+1$ elements in its boundary)
- The algorithm can be reduced to *Gaussian elimination*: $O(n^\omega)$ with $2 \leq \omega \leq 2.373$

The persistent homology pipeline



(S,d) finite metric

Vietoris Cech sublevel

Barcode Persist. diagram

Sample A

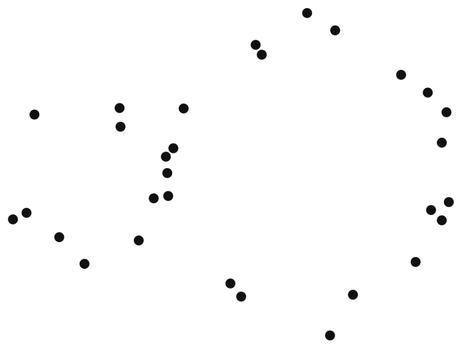
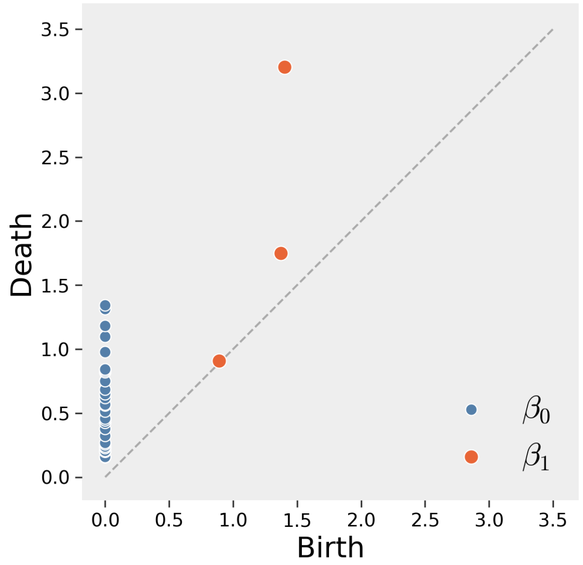


Diagram A



Sample B

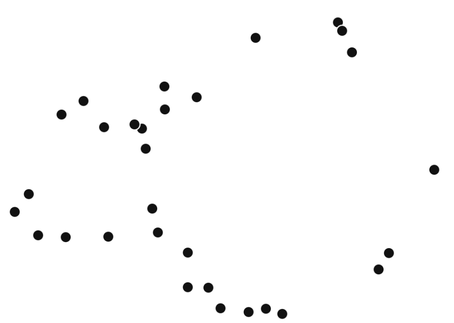
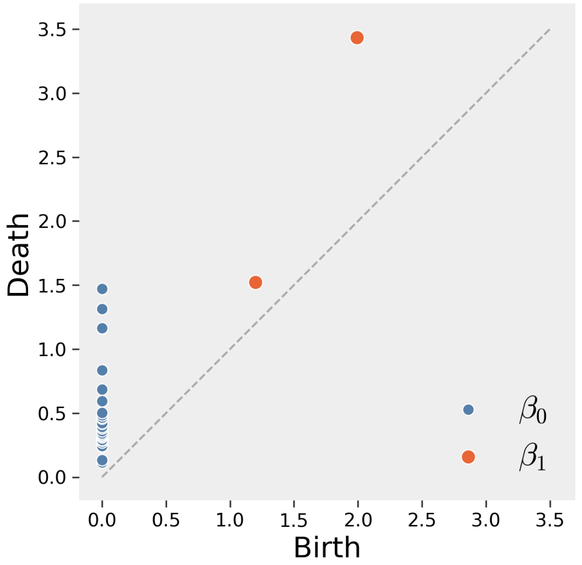


Diagram B



How can we compare diagrams?

Distances in diagrams

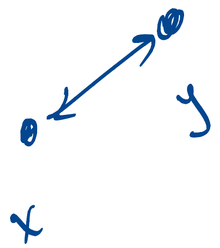
Lecture 8 — (More) persistent homology

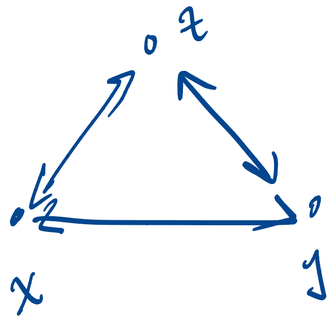
Distance functions

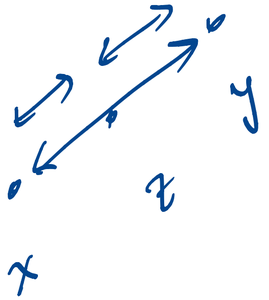
Def. Distance function

A distance function on a set X is a map $d : X \times X \to \mathbb{R}_{\geq 0}$ such that:

- **Positiveness**: $d(x, y) \geq 0$, $\forall x, y \in X$ and $d(x, y) = 0 \iff x = y$
- **Symmetry**: $d(x, y) = d(y, x)$, $\forall x, y \in X$
- **Triangle inequality:** $d(x, y) \leq d(x, z) + d(z, y)$, $\forall x, y, z \in X$

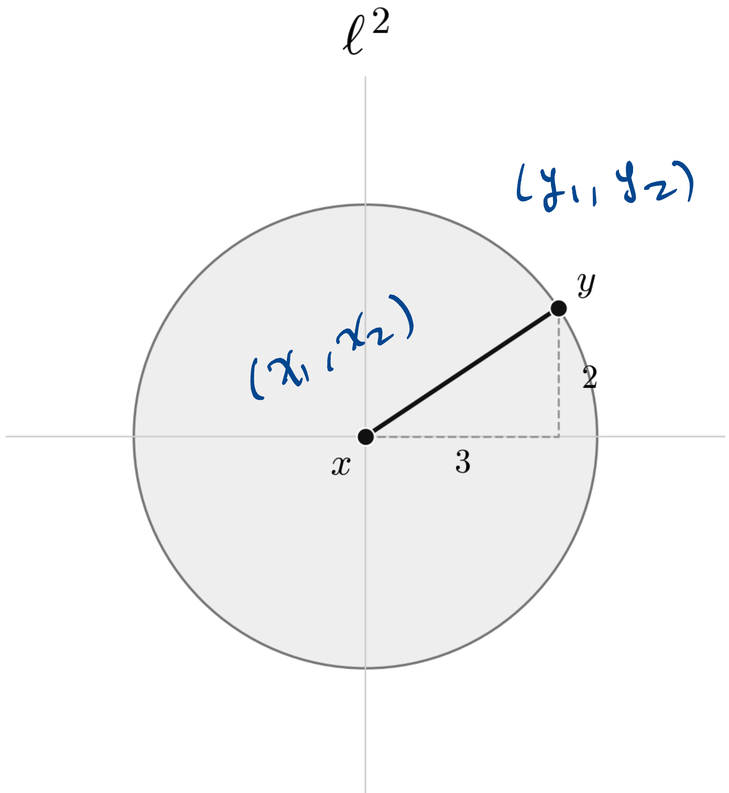






Euclidean distance in $\mathbb{R}^2$

$d : \mathbb{R}^2 \times \mathbb{R}^2 \longrightarrow \mathbb{R}_{\geq 0}$



$$d_2(x,y) = \sqrt{3^2 + 2^2} = \sqrt{13}$$

$$x = (0,0), \quad y = (3,2)$$

$B(x, \sqrt{13})$

Distance:

$$d(x,y) = \sqrt{|x_1 - y_1|^2 + |x_2 - y_2|^2}$$

Metric ball:

$$B(x,r) = \{y \in \mathbb{R}^2 : d(x,y) \leq r\}$$

In this case the distance comes from a norm:

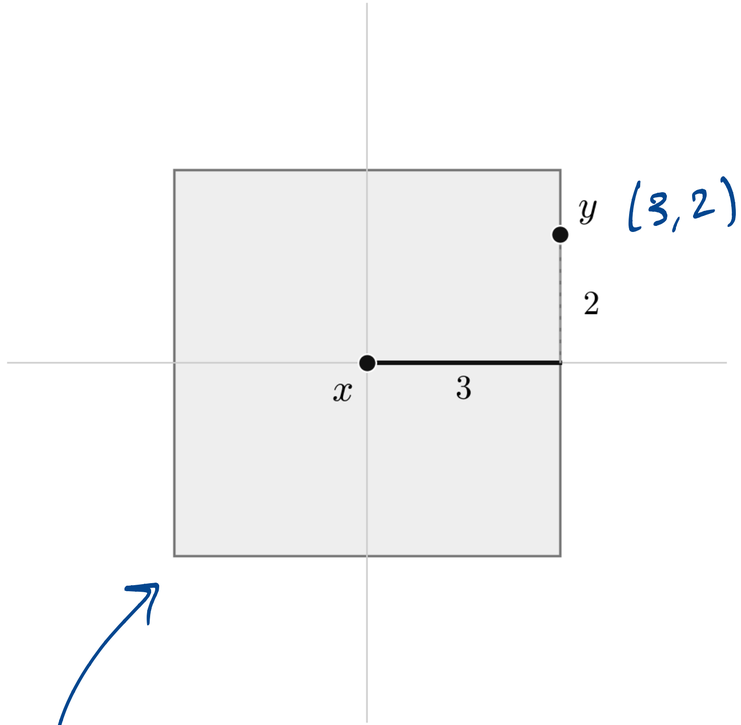
$$\|\cdot\|_2 : \mathbb{R}^2 \to \mathbb{R}_{\geq 0} \qquad \|x\|_2 = \sqrt{x_1^2 + x_2^2}$$

$$d(x,y) = \left(|x_1 - y_1|^2 + |x_2 - y_2|^2\right)^{1/2}$$

Other distances in $\mathbb{R}^2$

Manhattan distance

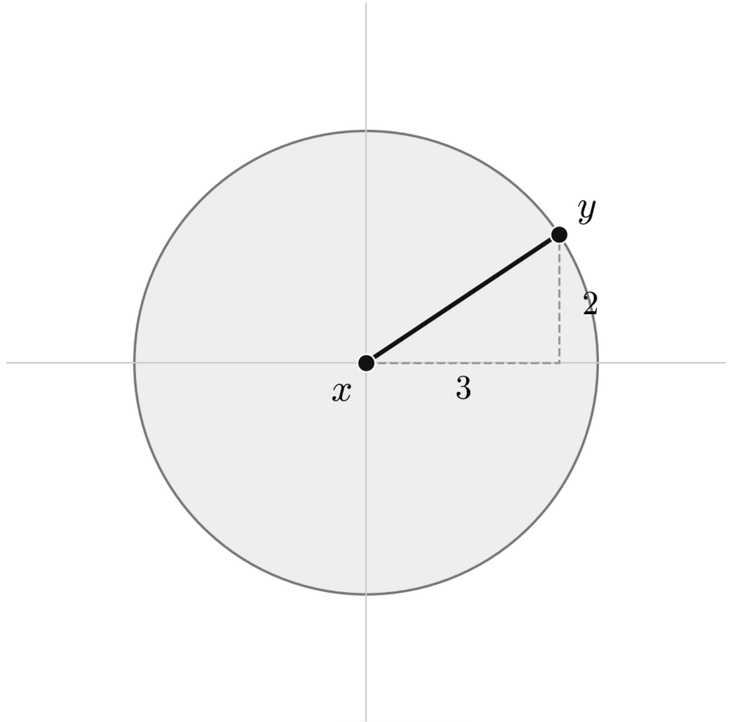
$$d_\infty(x,y) = \max\{|x_1 - y_1|, |x_2 - y_2|\}$$



$$d_\infty(x,y) = \max\{3,2\} = 3$$

$B_\infty(x, 3)$

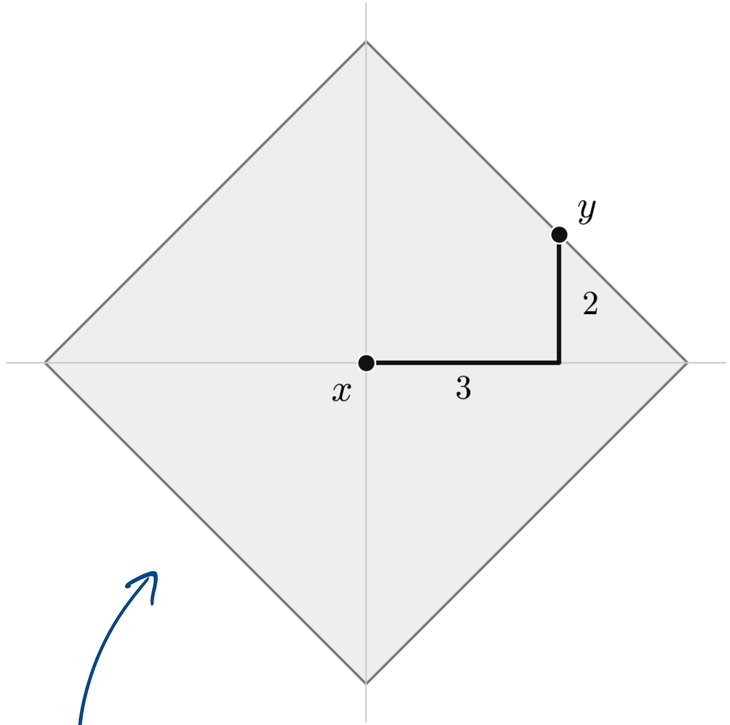
$$d_2(x,y) = \sqrt{|x_1 - y_1|^2 + |x_2 - y_2|^2}$$



$$d_2(x,y) = \sqrt{3^2 + 2^2} = \sqrt{13}$$

$$x = (0,0), \quad y = (3,2)$$

$$d_1(x,y) = |x_1 - y_1| + |x_2 - y_2|$$



$$d_1(x,y) = 3 + 2 = 5$$

$B_1(x, 5)$

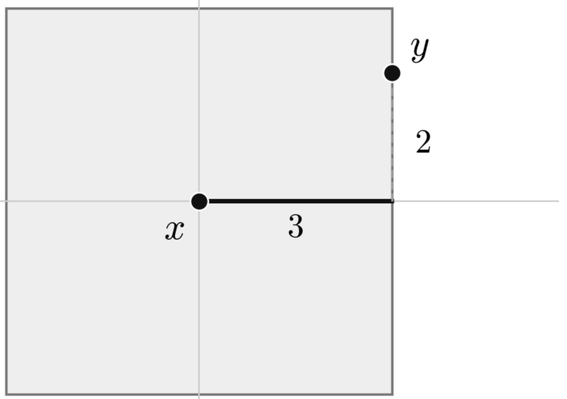
ℓ^p distances in $\mathbb{R}^2$

ℓ^p in $\mathbb{R}^d$: $d_p(x,y) = \left(\sum_{i=1}^{d} |x_i - y_i|^p\right)^{1/p}$

$x = (x_1, \ldots, x_d)$

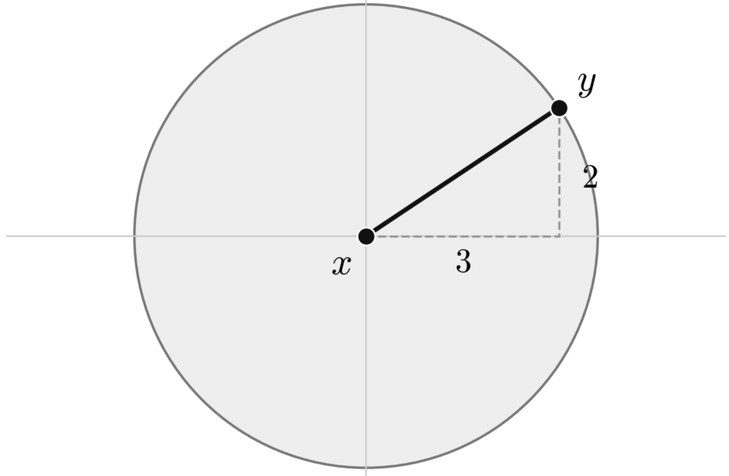
$y = (y_1, \ldots, y_d)$

ℓ^∞



$d_\infty(x,y) = \max\{3,2\} = 3$

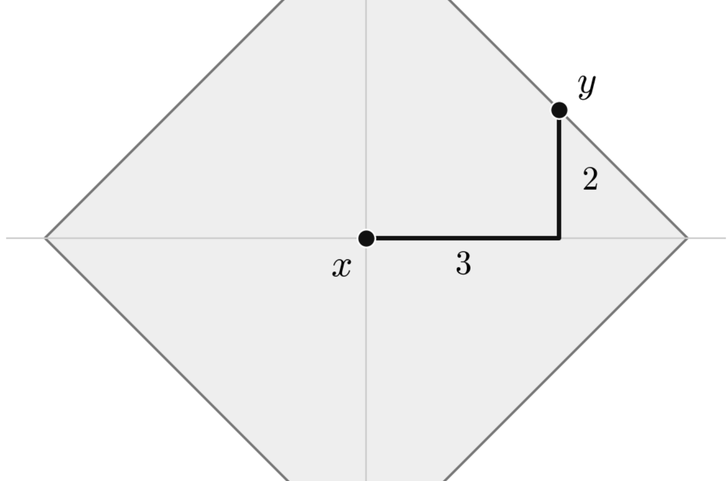
ℓ^2



$d_2(x,y) = \sqrt{3^2 + 2^2} = \sqrt{13}$

$x = (0,0), \quad y = (3,2)$

ℓ^1



$d_1(x,y) = 3 + 2 = 5$

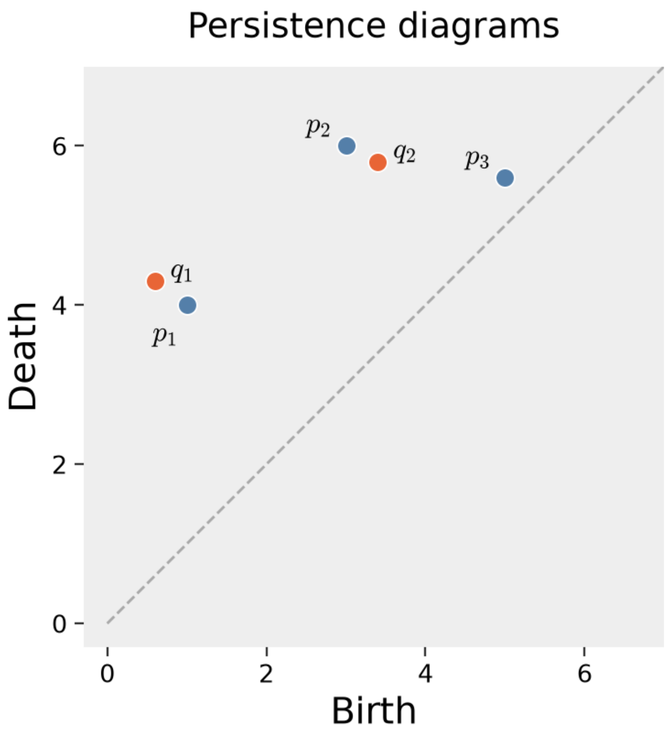
$$d_p(x,y) = \left(|x_1 - y_1|^p + |x_2 - y_2|^p\right)^{1/p}$$

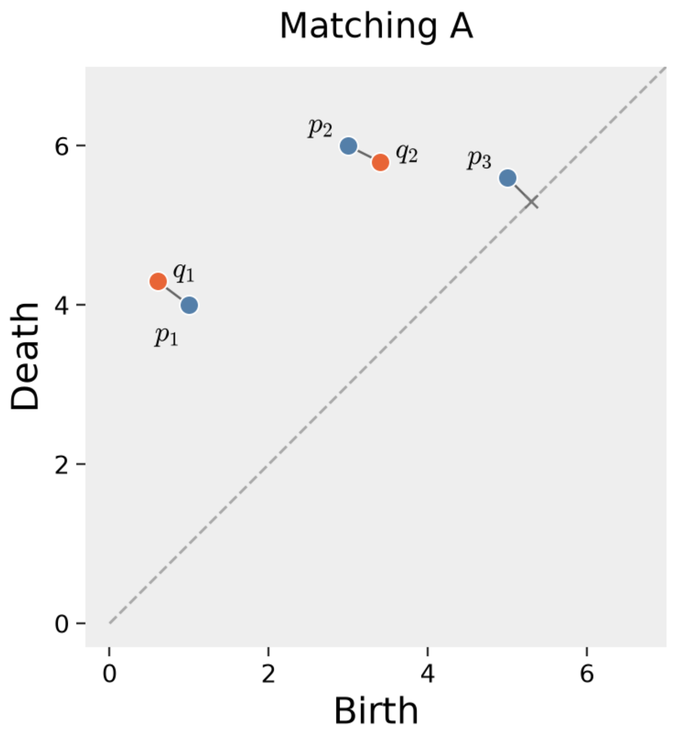
$$B_p(x,r) = \{y \in \mathbb{R}^2 : d_p(x,y) \le r\}$$

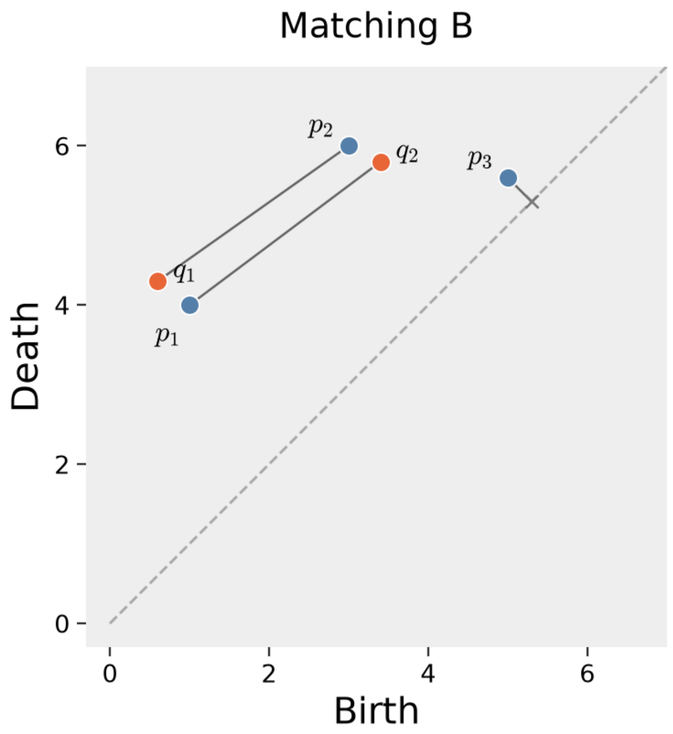
Matchings between diagrams

Def. Matching between persistence diagrams

A matching between persistence diagrams is a bijection $\psi : \mathrm{PD}(X) \to \mathrm{PD}(Y)$ between the points in the diagrams, where the points in the diagonal in both diagrams are counted with infinite multiplicity







Distances between diagrams

Def. Matching between persistence diagrams

A matching between persistence diagrams is a bijection $\psi : \mathrm{PD}(X) \to \mathrm{PD}(Y)$ between the points in the diagrams, where the points in the diagonal in both diagrams are counted with infinite multiplicity

Bottleneck distance

$$d_B\left(\mathrm{PD}(X), \mathrm{PD}(Y)\right) = \inf_{\psi:\mathrm{PD}(X)\to\mathrm{PD}(Y)} \max\{\|x - \psi(x)\|_\infty : \forall x \in \mathrm{PD}(X)\}$$

$d_\infty(x, \psi(x))$

Wasserstein distance

$$W_p\left(\mathrm{PD}(X), \mathrm{PD}(Y)\right) = \inf_{\psi:\mathrm{PD}(X)\to\mathrm{PD}(Y)} \left(\sum_{x\in\mathrm{PD}(X)} \|x - \psi(x)\|_p^p \right)^{1/p}$$

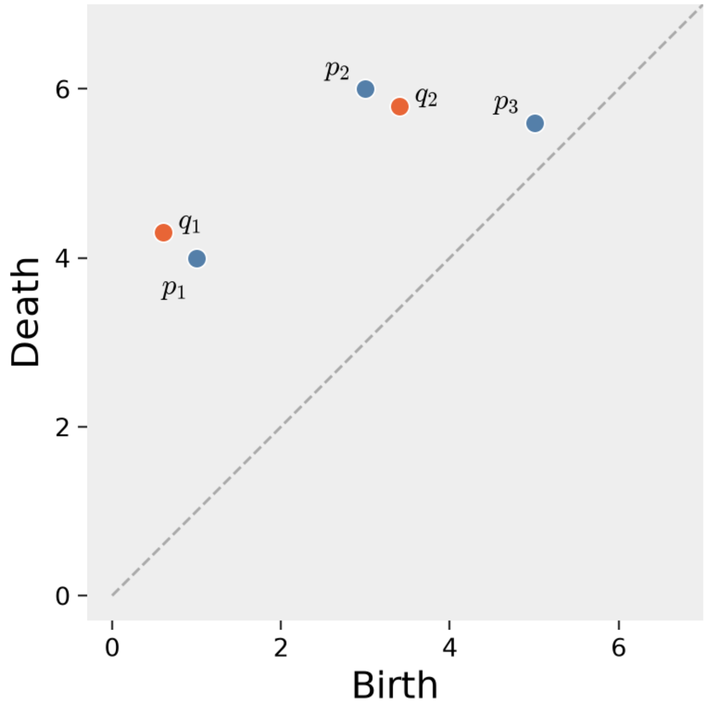
$W_{p,q}$

$d_p(x, \psi(x))$ p q

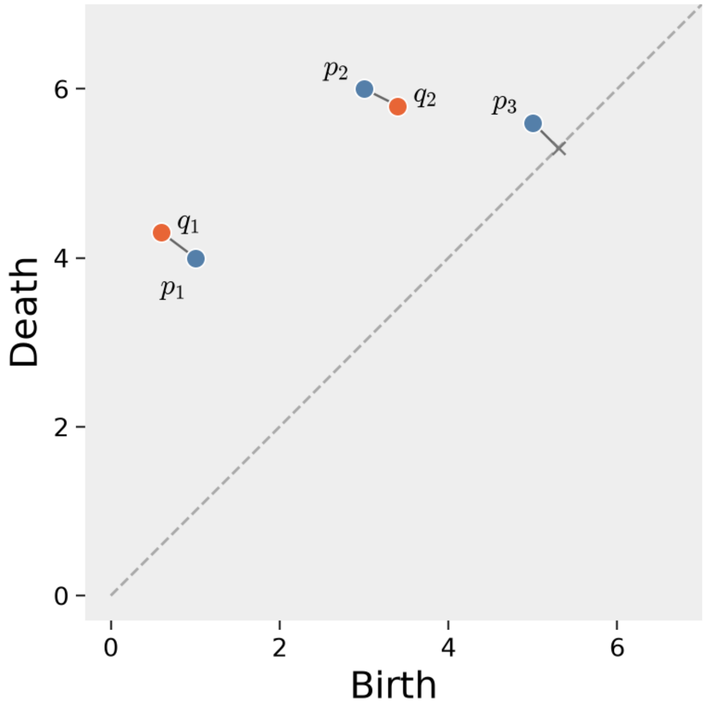
Distances between diagrams

In this case the matchings for the Bottleneck and the Wasserstein agree...

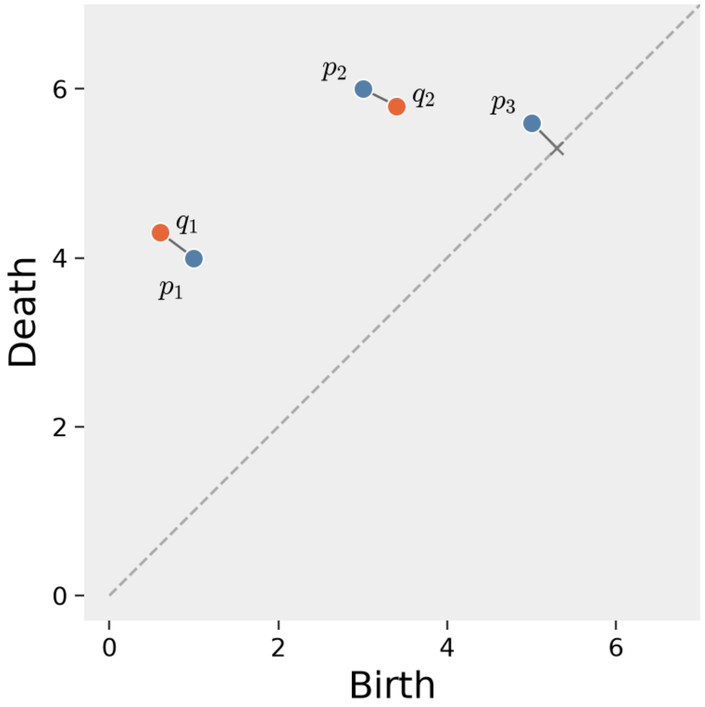
Persistence diagrams



Bottleneck optimum



W_2 optimum



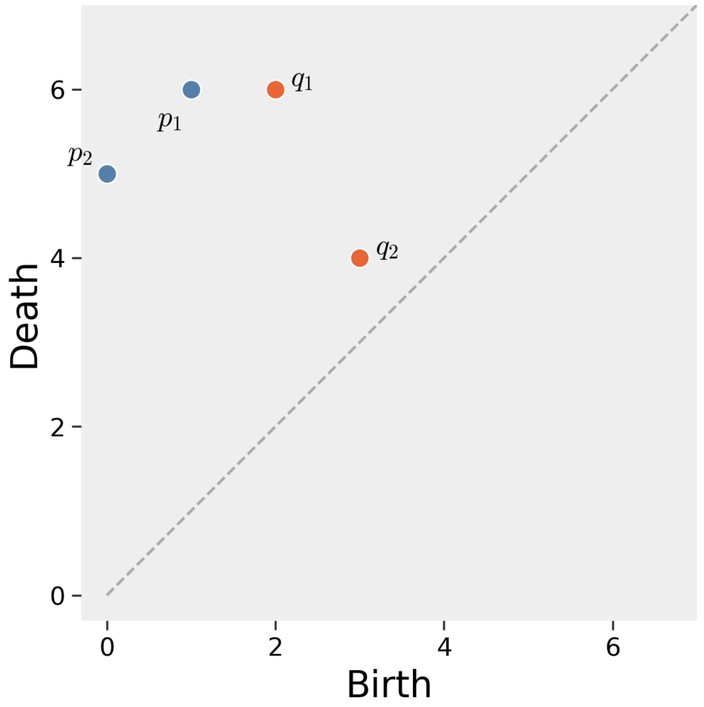
$$d_B = \max \|x - \psi(x)\|_\infty = 0.4$$

$$W_2 = \left(\sum \|x - \psi(x)\|_2^2\right)^{1/2} = 0.7937$$

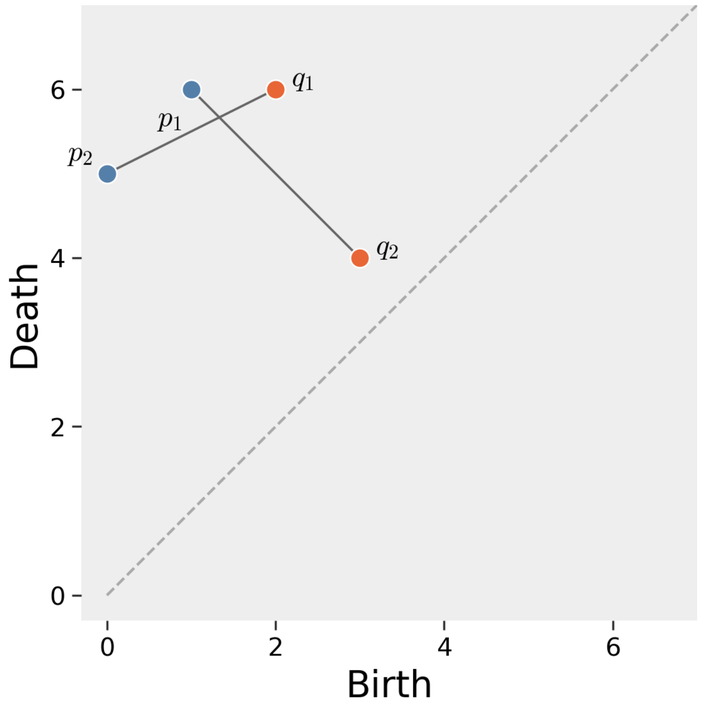
Distances between diagrams

But this needs not always be the case

Persistence diagrams

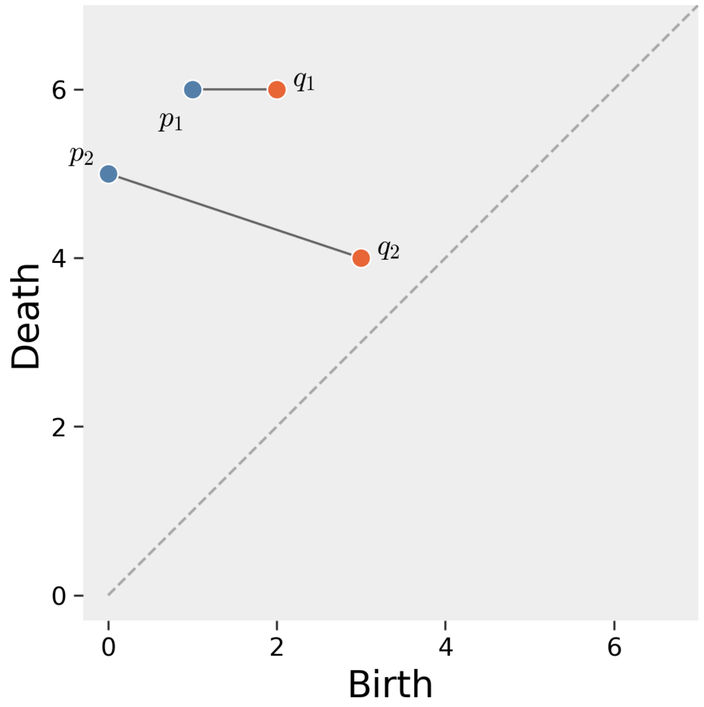


Bottleneck optimum



$$d_B = \max \|x - \psi(x)\|_\infty = 2$$

W_2 optimum



$$W_2 = \left(\sum \|x - \psi(x)\|_2^2\right)^{1/2} = 3.3166$$

Stability

Lecture 8 — (More) persistent homology

Stability in persistence

Input data space → $\mathscr{D}$; Space of invariants → $\mathscr{I}$

Recipe for a stability result in TDA: A given choice of invariant $\mathrm{inv} : \mathscr{D} \to \mathscr{I}$ is *stable* if it is Lipschitz

There exists $C > 0$ such that $\quad d_{\mathscr{I}}(\mathrm{inv}(X), \mathrm{inv}(Y)) \leq C \cdot d_{\mathscr{D}}(X, Y), \quad \forall X, Y \in \mathscr{D}$

So far we only have the barcodes as invariants of the persistence pipeline, so we want something like:

$$d_B(\mathrm{PD}(X), \mathrm{PD}(Y)) \leq C \cdot d(X, Y) \text{ for any two point clouds } X, Y \subseteq \mathbb{R}^d$$

How do we measure distances between point clouds??

Correspondence

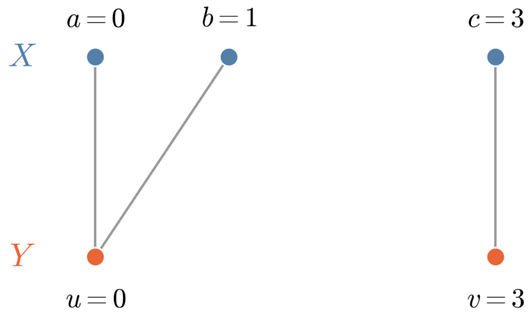
Def. Correspondence

Let $X, Y \subset \mathbb{R}^d$ be two point clouds. A correspondence is a subset of pairs $C \subset X \times Y$ such that:

- For all $x \in X$, there is some $y \in Y$ such that $(x, y) \in C$
- For all $y \in Y$, there is some $x \in X$ such that $(x, y) \in C$

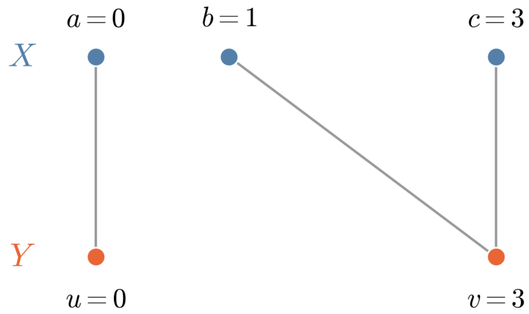
$X, Y \subseteq \mathbb{R}$

C_1



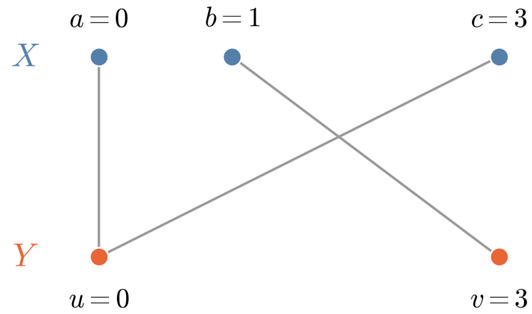
(a, u), (b, u), (c, v)

C_2



(a, u), (b, v), (c, v)

C_3



(a, u), (b, v), (c, u)

Distortion

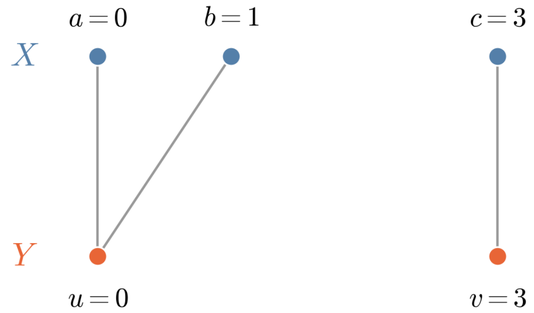
Def. Distortion

For $X, Y \subseteq \mathbb{R}^d$ finite point clouds and $C \subset X \times Y$ a correspondence

$$\operatorname{dis}(C) = \sup\{\, |d_X(x,x') - d_Y(y,y')| : (x,y),(x',y') \in C\}$$

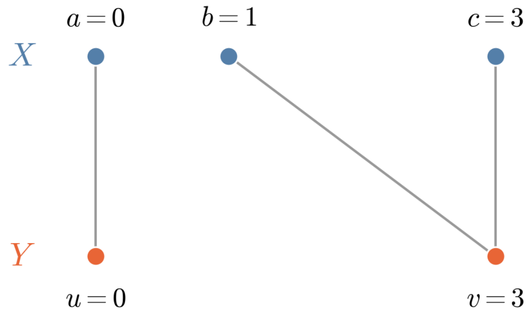
$X, Y \subseteq \mathbb{R}$

C_1



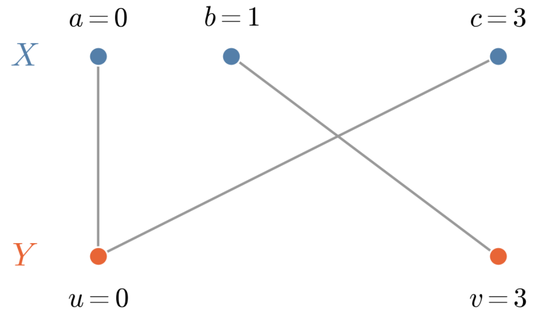
$\operatorname{dis}(C_1) = 1$

C_2



$\operatorname{dis}(C_2) = 2$

C_3



$\operatorname{dis}(C_3) = 3$

(a,u), (b,u): $||a-b| - |u-u|| = |1-0| = 1$

(b,u), (c,v): $||b-c| - |u-v|| = |2-3| = 1$

(a,u), (c,v): $||a-c| - |u-v|| = |3-3| = 0$

Gromov—Hausdorff distance

Def. Gromov—Hausdorff distance

Given two point clouds $X, Y \subseteq \mathbb{R}^d$, their

$$d_{GH}(X, Y) = \frac{1}{2} \min_{C \subset X \times Y} \operatorname{dis}(C)$$

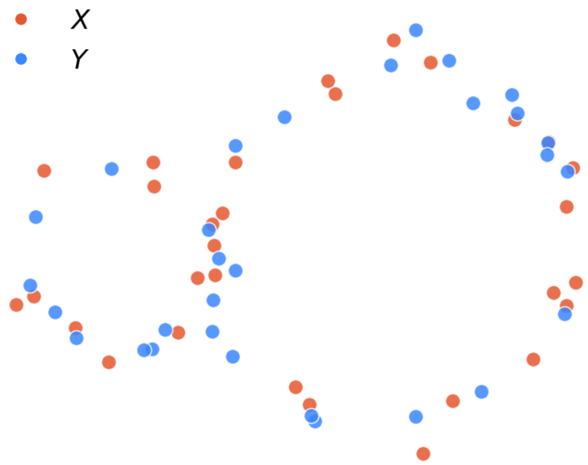
For the previous point cloud, the minimal distortion is actually 1, so $d_{GH}(X, Y) = \frac{1}{2}$

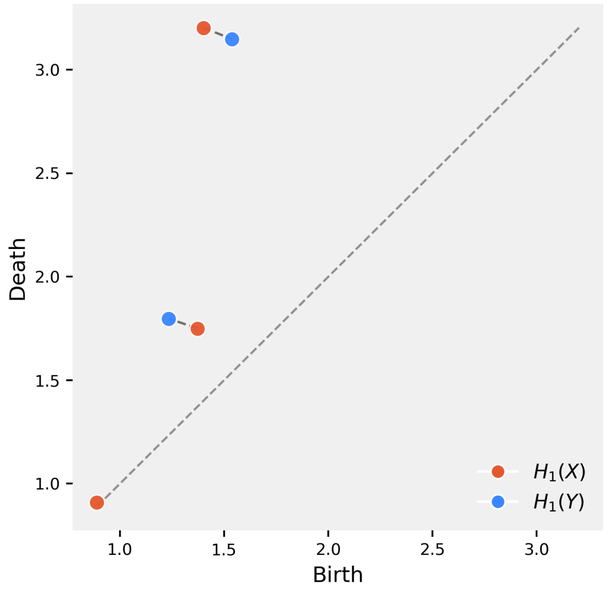
Bottleneck stability

Thm. Bottleneck stability of VR diagrams

If $X, Y \in \mathbb{R}^d$ are two point clouds and $\mathrm{PD}(X)$, $\mathrm{PD}(Y)$ are the persistence diagrams for their Vietoris-Rips filtrations, then we have:

$$d_B\left(\mathrm{PD}(X), \mathrm{PD}(Y)\right) \leq 2 \cdot d_{GH}(X, Y)$$





Chazal, Frédéric, Vin de Silva, and Steve Oudot. "Persistence Stability for Geometric Complexes." *Geometriae Dedicata* 173, no. 1 (2014): 193–214.

Vectorizations

Lecture 8 — (More) persistent homology

The persistent homology pipeline

Barcode + vectorization } Betti curves, Pers. Landscapes.



Now we have barcodes/diagrams and we have distance between barcodes/diagrams, that is, we have turned the space of barcodes/diagrams into a *metric space*

It has been proven that this metric space is *amenable* to probability and statistics, that is, we can develop statistical methods for barcodes

However, because of the *geometric* properties of the space of barcodes, developing these statistical methods is highly non-trivial and many of the most basic assumption fail (for instance, means are not unique)

In addition, many data analysis or ML methods expect your input data to come from an Euclidean or an inner product space, not some generic metric space...

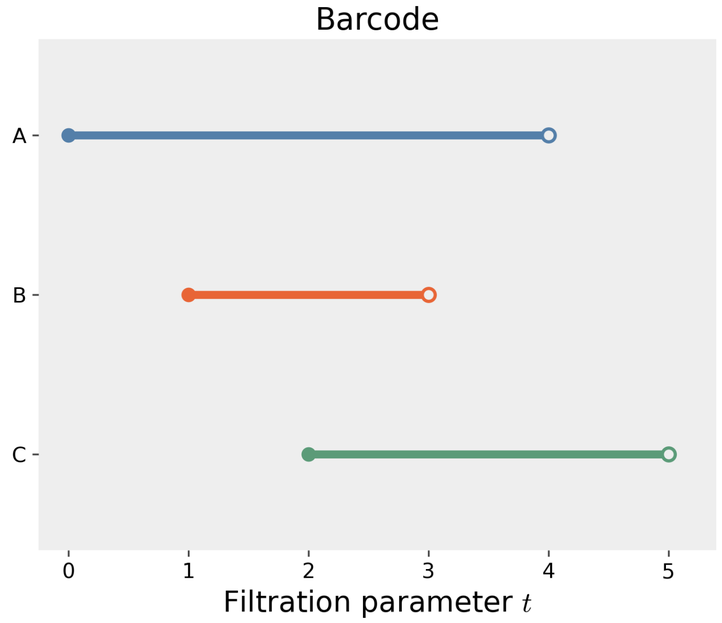
A possible solution for all these problems is to **vectorize** your barcode/diagram, that is, turn it into a feature vector or a function that has nicer statistical properties

Statistical summary of the barcode

Define **numerical descriptors** for your barcode:

- Births (vector)
- Deaths (vector)
- Persistences (vector)
- Ratios birth/death (vector)
- Number of bars in the diagram (scalar)
- Total persistence (scalar)
- Persistent entropy[1] (scalar)

For the vectorial measures, pick some statistics (mean, standard deviation, median, quartiles...) and build a **feature vector** collecting their values for the empirical distributions of these **statistical descriptors of the barcode**



Statistical summaries

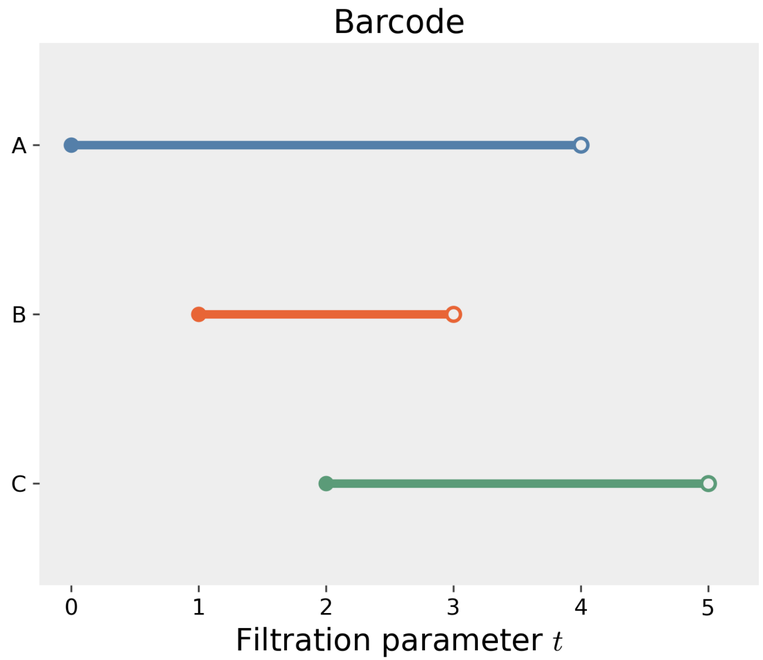
Number of bars	3
Mean birth	1
Mean death	4
Mean lifetime	3
Maximum lifetime	4
Total lifetime	9

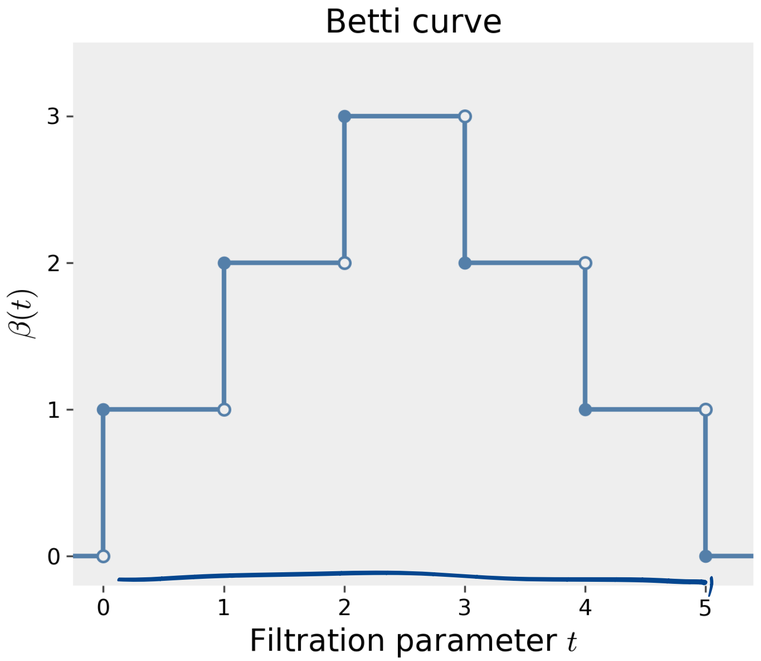
$$s(D) = (3, 1, 4, 3, 4, 9)$$

[1] Chintakunta, H., Gentimis, T., Gonzalez-Diaz, R., Jimenez, M. J., and Krim, H. "An Entropy-Based Persistence Barcode." *Pattern Recognition* 48, no. 2 (2015): 391–401.

Betti curve

[1, 1, 2, 2, 3, 3, ...





Def. Betti curve

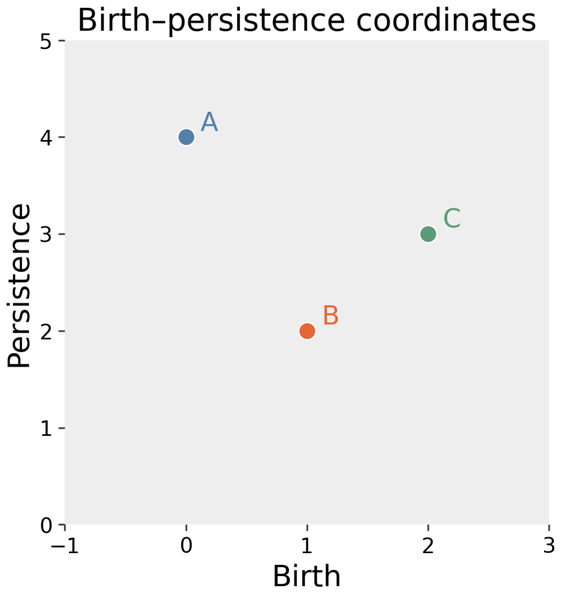
Given a persistent homology module $\{H_k(K_t) : t \in \mathbb{R}\}$, the Betti curve is a function $\beta_k : \mathbb{R} \to \mathbb{N}$ defined as

$$\beta_k(t) = \beta_k(K_t) = \#\{\text{alive bars in } H_k(K_t)\}$$

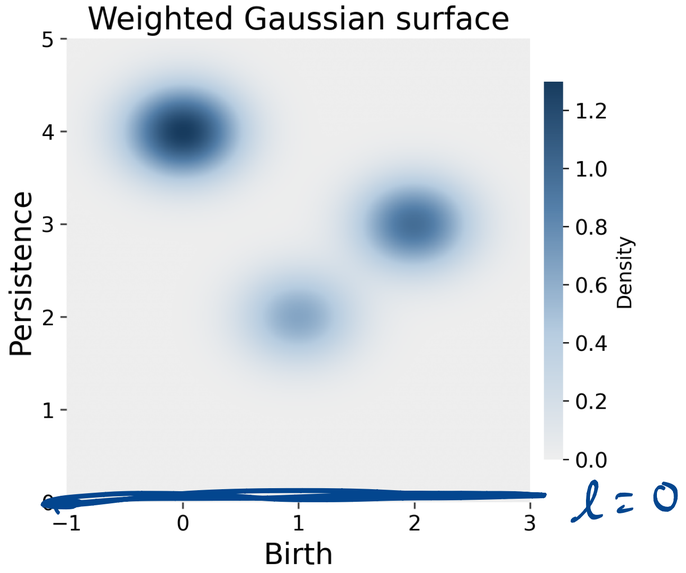
→ discretize it using different resolutions

Giusti, Chad, Eva Pastalkova, Carina Curto, and Vladimir Itskov. "Clique Topology Reveals Intrinsic Geometric Structure in Neural Correlations." *Proceedings of the National Academy of Sciences* 112, no. 44 (2015): 13455–60.

Persistence image

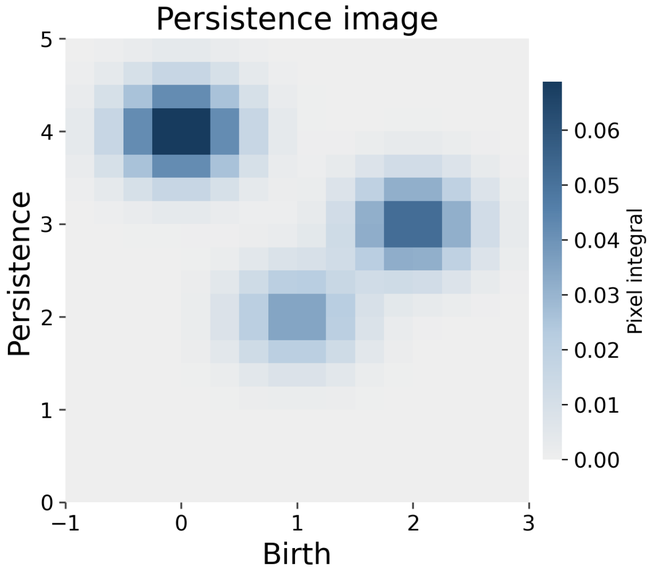


$(b, d) \mapsto (b, d - b)$



$\sigma = 0.35, \quad w(\ell) = \min(\ell/4, 1)$

$w(b, \ell)$ //

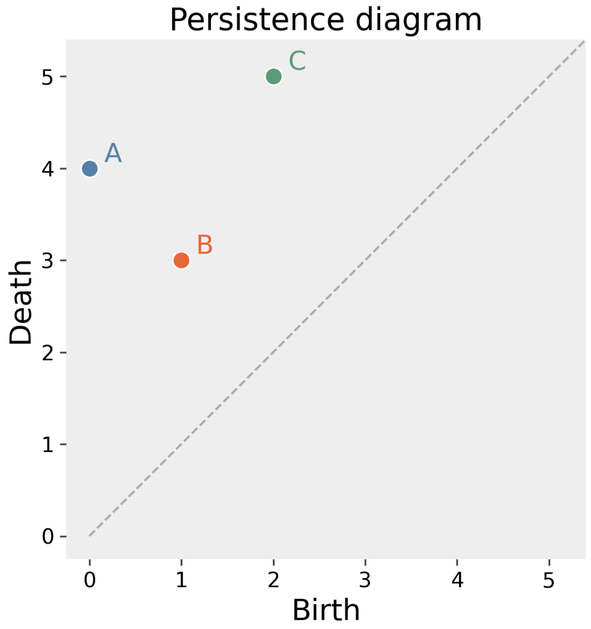


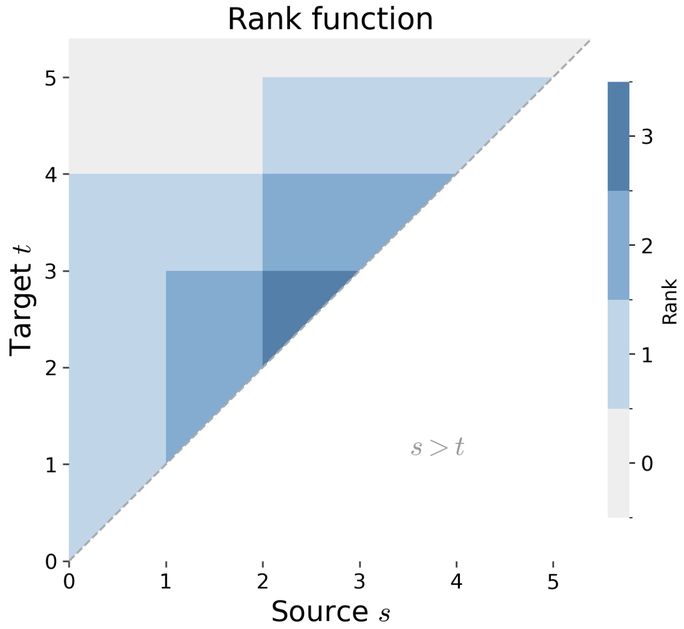
20×16 pixels $\longrightarrow \mathbb{R}^{320}$

1. Change to *persistence coordinates* in the persistence diagram $(b, d) \mapsto (b, l = b - d)$
2. Let $\phi_u : \mathbb{R}^2 \to \mathbb{R}$ be a differentiable probability distribution with mean $u = (u_b, u_d) \in \mathbb{R}^2$
3. Let $w : \mathbb{R}^2 \to \mathbb{R}$ be a non-negative weighting function, that is 0 in the horizontal axis, continuous and piece-wise differentiable.
   The **persistence surface** is then: $\rho(b, d) = \sum_{u \in \mathrm{PD}(X)} w(u)\, \phi_u(b, d)$ $\quad \rho : \mathbb{R}^2 \longrightarrow \mathbb{R}$
4. Divide the plane into pixels and integrate to obtain the **persistence image**

Adams, Henry, Tegan Emerson, Michael Kirby, et al. "Persistence Images: A Stable Vector Representation of Persistent Homology." *Journal of Machine Learning Research* 18, no. 8 (2017): 1–35.

Rank function





<u>Def.</u> Rank function

Given a persistent homology module $\{H_k(K_t) : t \in \mathbb{R}\}$, its rank function is defined as:

$$\mathrm{rk} : \mathbb{R}^2 \to \mathbb{R}, \quad \mathrm{rk}(s,t) = \begin{cases} \dim \mathrm{im}\left(H_k(K_s) \to H_k(K_t)\right) & \text{if } s \le t, \\ 0 & \text{otherwise}. \end{cases}$$

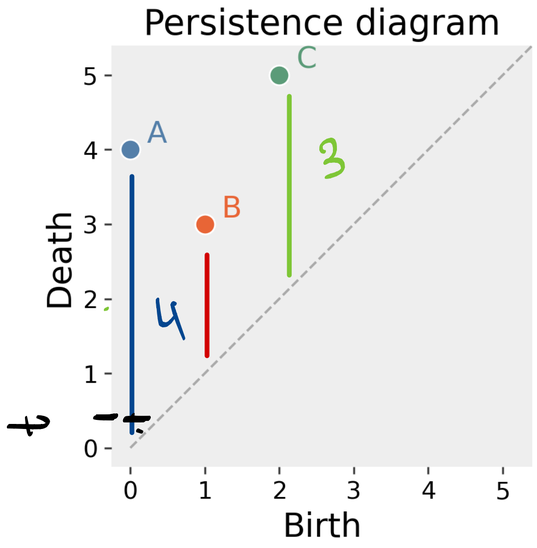
This is the same as counting the number of bars at s that stay alive at t

$$\mathrm{rk}(s,t) = \#\{i : b_i \le s \le t < d_i\}$$

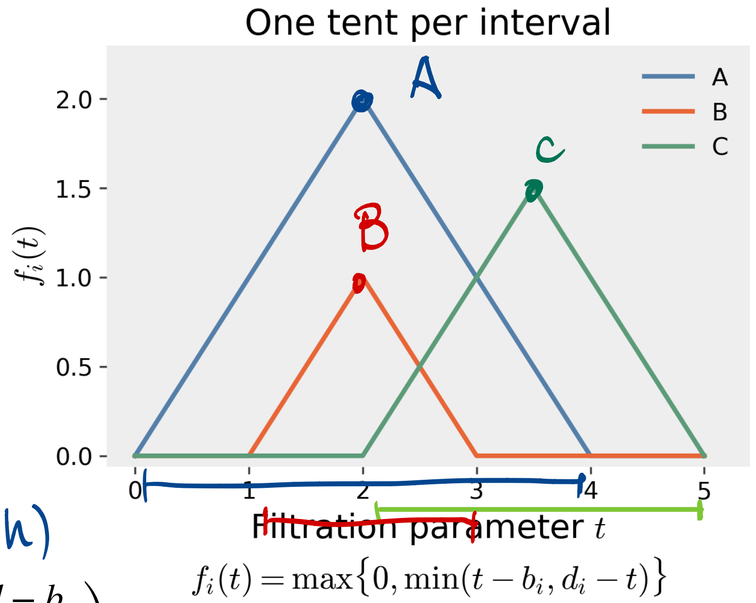
Frosini, Patrizio. A Distance for Similarity Classes of Submanifolds of a Euclidean Space. *Bulletin of the Australian Mathematical Society* 42, no. 3 (1990): 407–15.

Frosini, Patrizio. Discrete Computation of Size Functions. *Journal of Combinatorics, Information & System Sciences* 17, nos. 3–4 (1992): 232–50.

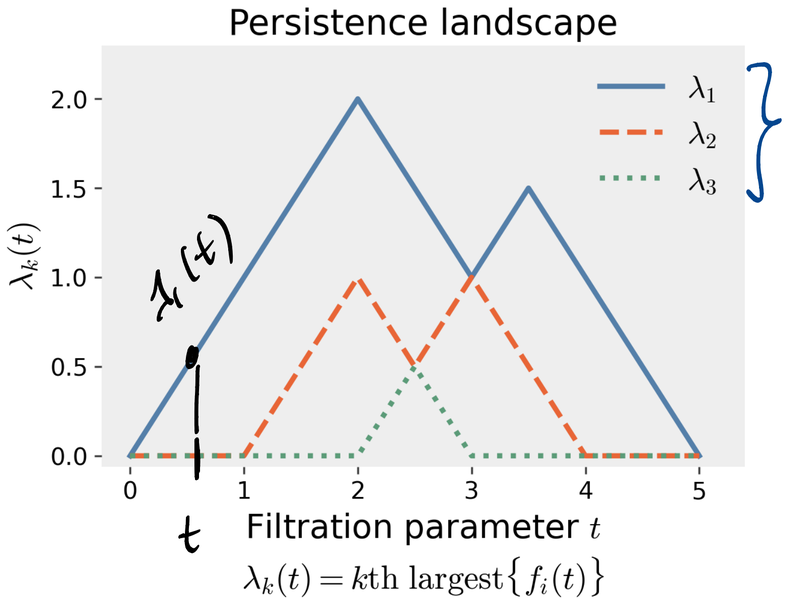
Persistence landscape



$$(b,d) \mapsto \left(\frac{b+d}{2}, \frac{d-b}{2}\right)$$



$$f_i(t) = \max\{0, \min(t - b_i, d_i - t)\}$$



$$\lambda_k(t) = k\text{th largest}\{f_i(t)\}$$

<u>Def.</u> Persistence landscape

Given a persistent homology module $\{H_k(K_t) : t \in \mathbb{R}\}$, its persistence landscape is defined as:

$$\lambda_k : \mathbb{R} \to \mathbb{R} \cup \{\pm\infty\}, \quad \lambda_k(t) = \sup\{r \geq 0 : \mathrm{rk}\,(t-r, t+r) \geq k\} = k\text{th largest } \{f_i(t)\}_i$$

Bubenik, Peter. Statistical Topological Data Analysis Using Persistence Landscapes. *Journal of Machine Learning Research* 16, no. 3 (2015): 77–102.

To sum up

Today we have learnt how to...

- How to compute barcodes
- How to compute distances between barcodes
- How to vectorize barcodes to use them as input in ML models

Next time we will see how to incorporate persistence into learning!

Thank you for your attention!



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Lecture 8 — (More) Persistent Homology

Inés García Redondo — November 3, 2026